BHANU TEJA GULLAPALLI

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PROFESSIONAL SUMMARY

AI engineer and researcher building production machine learning systems for healthcare and wearable sensing applications. Experienced in online learning, reinforcement learning, Bayesian modeling, LLM-enabled workflows, and end-to-end AI deployment, including APIs, monitoring, evaluation, and operational reliability. Led AI systems deployed in active randomized controlled trials serving real-world participants.

TECHNICAL SKILLS

AI & Machine Learning: Reinforcement Learning, Online Learning, Bayesian Modeling, Survival Analysis, Sequential Decision Making, Time-Series Modeling, Deep Learning, Uncertainty Quantification, Transformers, Gaussian Processes

Production AI Systems: Production ML Systems, REST APIs, Docker, PostgreSQL, Monitoring & Alerting, Logging, Load Testing, Model Deployment, Model Retraining Workflows, Reproducibility Frameworks

AI Development Tools: Large Language Models (LLMs), Agentic Coding Workflows, LLM-Assisted Development

Frameworks & Tools: Python, PyTorch, Scikit-learn, TensorFlow, HuggingFace, Pandas, NumPy, PySpark, Linux, Git, Bash

EXPERIENCE

Postdoctoral Researcher, Harvard, Boston, MA — July 24 - Current

- Developed personalized intervention-timing algorithms using Bayesian learning, survival modeling, and uncertainty-aware decision making, improving intervention-timing evaluation performance by 15%.
- Led deployment of a personalized AI intervention platform in an active 135-day randomized controlled trial, supporting real-world participant interactions, longitudinal data collection, and online decision-making.
- Designed backend APIs, databases, Dockerized services, and model retraining workflows supporting intervention delivery with approximately 96% operational reliability.
- Built monitoring, alerting, logging, fallback, validation, and recovery systems, limiting production issues to an average of 7 of 135 study days per participant while enabling recovery of affected data.
- Developed reproducibility frameworks for online learning systems; reproduced intervention-timing performance across offline testing and live deployment, co-authored a peer-reviewed workflow paper on reproducible online AI in digital health, and led adoption of agentic AI coding workflows across the lab.

Research Intern, Optum AI labs — June 23 -September 23

- Developed continuous risk-estimation models for Type 2 Diabetes Mellitus using multimodal continuous glucose monitoring data to support personalized mobile health interventions.

[1] Use URL bhanutejagullapalli.github.io in case hyperlinks don't work

- Built passive sensing and data-processing pipelines for real-time behavioral and physiological risk modeling.
- Designed uncertainty-aware intervention strategies using probabilistic risk scores, temporal behavior patterns, and sampling-based decision methods.

Research Intern, Samsung Research America, Mountain View, CA May 22 - September 22

- Designed signal-processing and machine learning pipelines to extract heart-rate variability features from wearable earbud PPG sensors.
- Developed stress-detection models validated on human-subject data collected during standardized stressor tasks.
- Improved robustness of physiological feature extraction under noisy wearable sensing conditions.

Software Engineer, Samsung R&D Institute, Bangalore, India July 15 - December 16

- Shipped multimedia features for Samsung Galaxy video editing applications, including highlight generation, slow-motion editing, automated story generation, and theme-based video editing workflows.
- Collaborated with mobile systems, design, and UI teams to optimize performance and user experience for production Android devices, including Samsung Galaxy S8 Video Editor Pro.

SELECTED PUBLICATIONS

- Reproducible workflow for online AI in digital health
  *Ghosh, S.,**Gullapalli, B.T.**, Gao, D., Gazi, A.H., Trella, A., and Murphy, S.A*
  Philosophical Transactions of the Royal Society 2026
- SigmaScheduling: Uncertainty-Informed Scheduling of Decision Points for Intelligent Mobile Health Interventions
  *Gazi, A.H., **Gullapalli, B.T.**, Gao, D., Marlin, B.M., Shetty, V., and Murphy, S.A*
  BSN 2025
- Opioid misuse detection from cognitive and physiological data with temporal fusion deep learning
  ***Gullapalli, B.T.**, Luo, Y., Rahman, T., and Garland, E.L*
  Drug and Alcohol Dependence 2025
- Pharmacokinetics-Informed Neural Network for Predicting Opioid Administration Moments with Wearable Sensors
  ***Gullapalli, B.T.**, Garland, E.L., Carreiro, S., Chapman, B. P., and Rahman, T*
  IAAI 2024

EDUCATION

Ph.D. in Data Science, University of California, San Diego September 18 - June 24
Dissertation focused on wearable sensing, digital biomarkers, and deep learning models for detecting stages of the addiction cycle, including craving, drug use, and euphoria. Supported by NSF and Optum Labs grants.
Advised by *Prof. Tauhidur Rahman*

M.Sc. in Computer Science, University of Massachusetts, Amherst Feb 17 - Sept 18

B.Sc. in Computer Science, Indian Institute of Technology, Guwahati July 11 - May 15